Customer Churn Prediction — Machine Learning Approach Report

1. Introduction

Customer churn prediction is a critical use case in financial services, enabling organizations to identify customers at risk of leaving and implement proactive retention strategies. Early identification of churn risk can help reduce revenue loss, improve customer satisfaction, and optimize marketing resource allocation.

The objective of this project was to develop and evaluate machine learning models to identify customers likely to churn using demographic, transactional, and behavioral engagement data.

The analysis focused on:

- Preventing data leakage using pipelines
- Handling class imbalance appropriately
- Evaluating model stability through cross-validation
- Comparing models using ROC-AUC
- Understanding business trade-offs between precision and recall

The goal was to determine whether meaningful predictive signal exists and identify the most effective modeling approach.

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2. Dataset Description

The dataset consisted of customer-level information merged from multiple sources, including demographic attributes, transaction behavior, and service interaction metrics.

The final dataset contained approximately:

- **1000 customer records**
- **13 predictor features**
- **1 binary target variable (ChurnStatus)**

Feature Categories

Category	Example Features
Demographics	Age, Gender, Marital Status, Income Level
Engagement	Login Frequency, Days Since Last Login
Transactions	Total Transactions, Total Amount Spent, Avg Amount Spent

Service Usage	Banking Channel Used
Customer Support	Total Complaints

The target variable indicates whether a customer churned (1) or remained (0).

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3. Data Preparation and Preprocessing

To ensure robust and unbiased model evaluation, preprocessing steps were implemented using machine learning pipelines.

The preprocessing included:

- **Numerical features:** Standardization using StandardScaler
- **Categorical features:** One-Hot Encoding
- **Train-test split:** Stratified sampling (80/20) to preserve class distribution

All preprocessing transformations were performed within cross-validation folds to prevent data leakage.

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4. Handling Class Imbalance

The dataset exhibited moderate class imbalance:

- ~80% retained customers
- ~20% churned customers

Two strategies were used:

1. Class weighting (Logistic Regression)
2. SMOTE (Synthetic Minority Oversampling Technique) inside pipelines

SMOTE was applied only within training folds to avoid leakage from synthetic samples into evaluation data.

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5. Baseline Model — Dummy Classifier

A dummy classifier predicting the majority class was used to establish a baseline performance level.

This baseline confirms whether machine learning models capture meaningful predictive signal beyond random guessing.

Baseline Performance

Metric	Value
ROC-AUC	0.50

The baseline ROC-AUC of 0.50 represents random classification performance.

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6. Model 1 — Logistic Regression

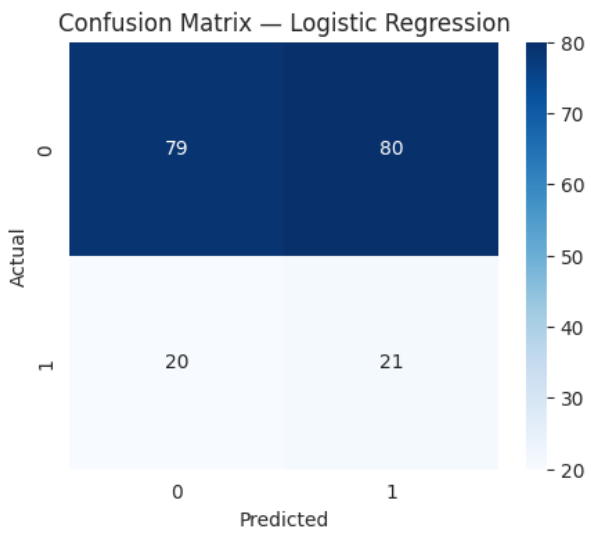
Logistic Regression was implemented as an interpretable baseline model with class weighting to address imbalance.

The model provides insight into whether linear relationships exist between features and churn behavior.

Logistic Regression Results

Metric	Value
CV ROC-AUC	0.52
Test ROC-AUC	0.48
Recall (Churn)	0.51
Precision (Churn)	0.21

The model achieved modest recall but low precision, indicating many false positives and weak separation between churned and retained customers.



7. Model 2 — Random Forest

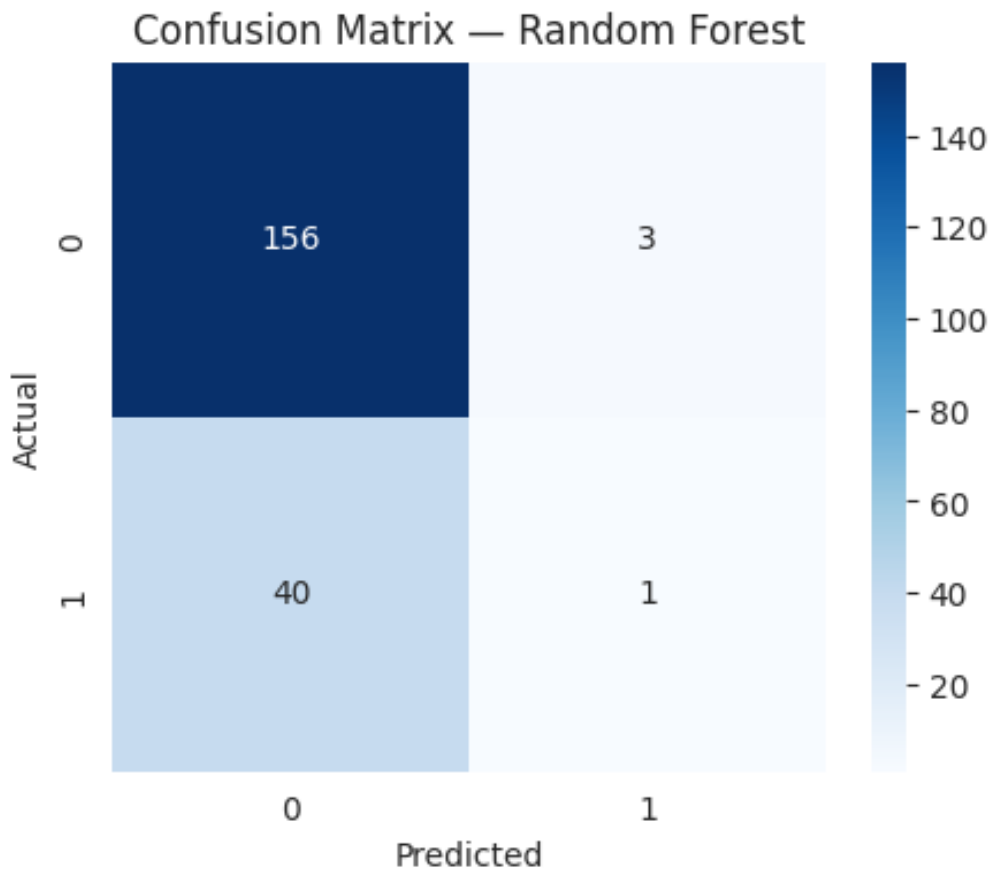
Random Forest was implemented to capture non-linear relationships and feature interactions.

Tree-based models are commonly effective for customer behavior prediction problems.

Random Forest Results

Metric	Value
CV ROC-AUC	0.54
Test ROC-AUC	0.49
Recall (Churn)	0.02
Precision (Churn)	0.25

The model achieved higher accuracy but extremely low churn recall, indicating strong bias toward the majority class.



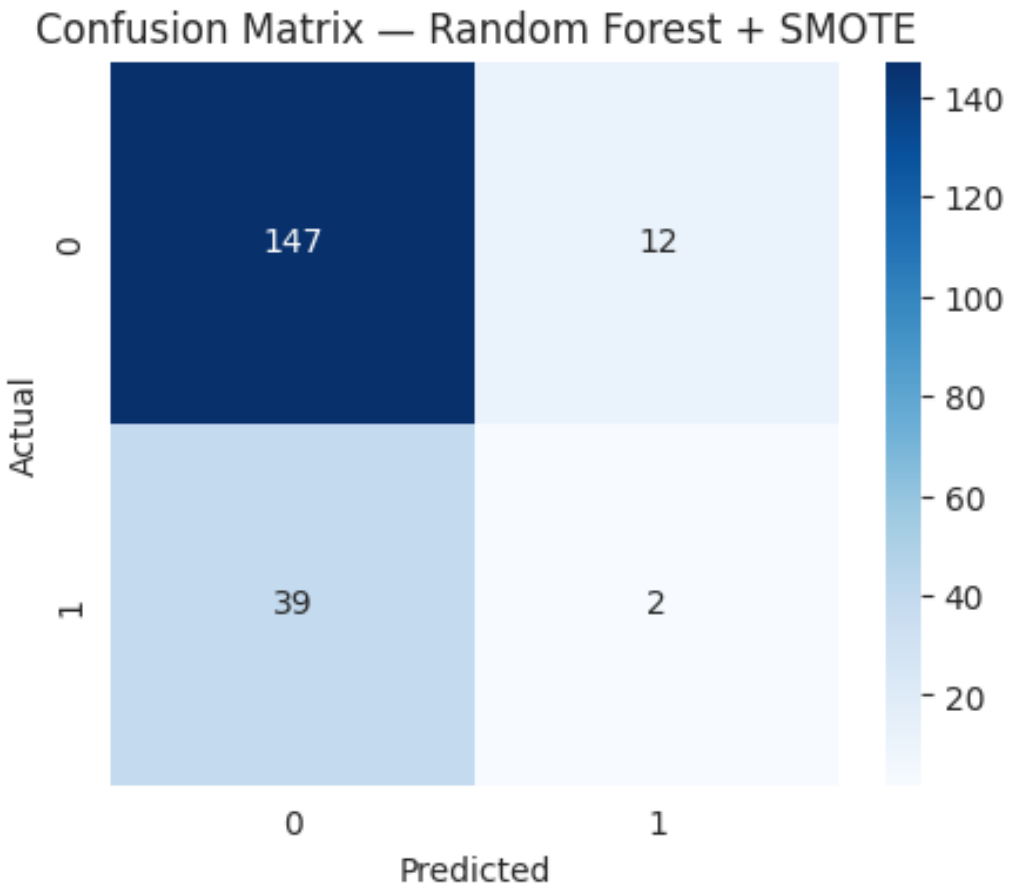
8. Model 3 — Random Forest with SMOTE

SMOTE oversampling was applied to improve sensitivity toward the minority class.

Random Forest + SMOTE Results

Metric	Value
CV ROC-AUC	0.53
Test ROC-AUC	0.49
Recall (Churn)	0.05
Precision (Churn)	0.14

SMOTE provided only marginal improvement, suggesting that class imbalance was not the primary performance constraint.



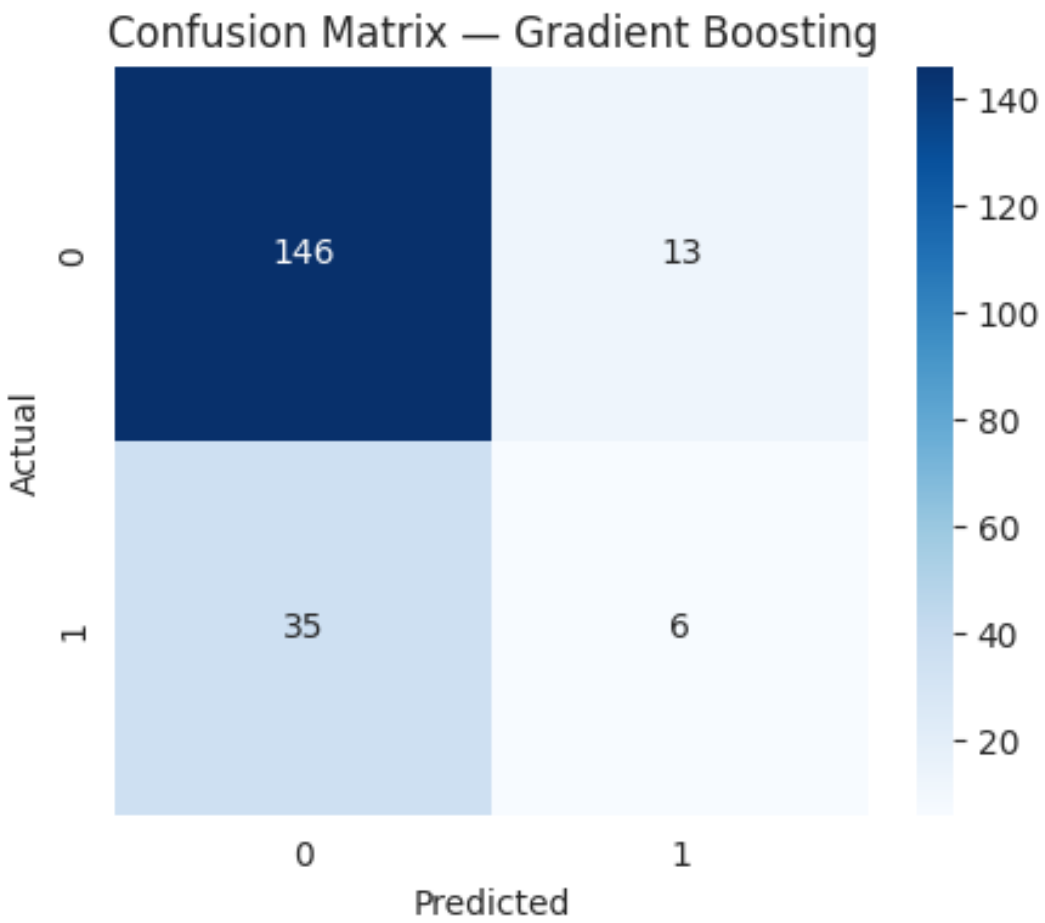
9. Model 4 — Gradient Boosting (Baseline)

Gradient Boosting was implemented as a more advanced ensemble technique capable of capturing subtle feature interactions.

Gradient Boosting Results

Metric	Value
CV ROC-AUC	0.55
Test ROC-AUC	0.48
Recall (Churn)	0.15
Precision (Churn)	0.32

This model demonstrated the strongest performance among baseline models.



10. Hyperparameter Optimization

Hyperparameters for Gradient Boosting were tuned using GridSearchCV with stratified cross-validation.

The optimal configuration was:

- Number of estimators: 200
- Learning rate: 0.05
- Maximum tree depth: 4

Tuned Model Performance

Metric	Value
Best CV ROC-AUC	0.573
Std Deviation	0.067

The tuned model achieved the highest predictive performance across all approaches.

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11. Model Comparison

Models were compared using cross-validated ROC-AUC.

Final Model Comparison Table

Model	*Mean ROC-AUC*
Dummy	0.50
Logistic Regression	0.52
Random Forest	0.54
Random Forest + SMOTE	0.53
Gradient Boosting	0.55
Gradient Boosting (Tuned)	**0.57**

Gradient Boosting with hyperparameter tuning achieved the best performance.

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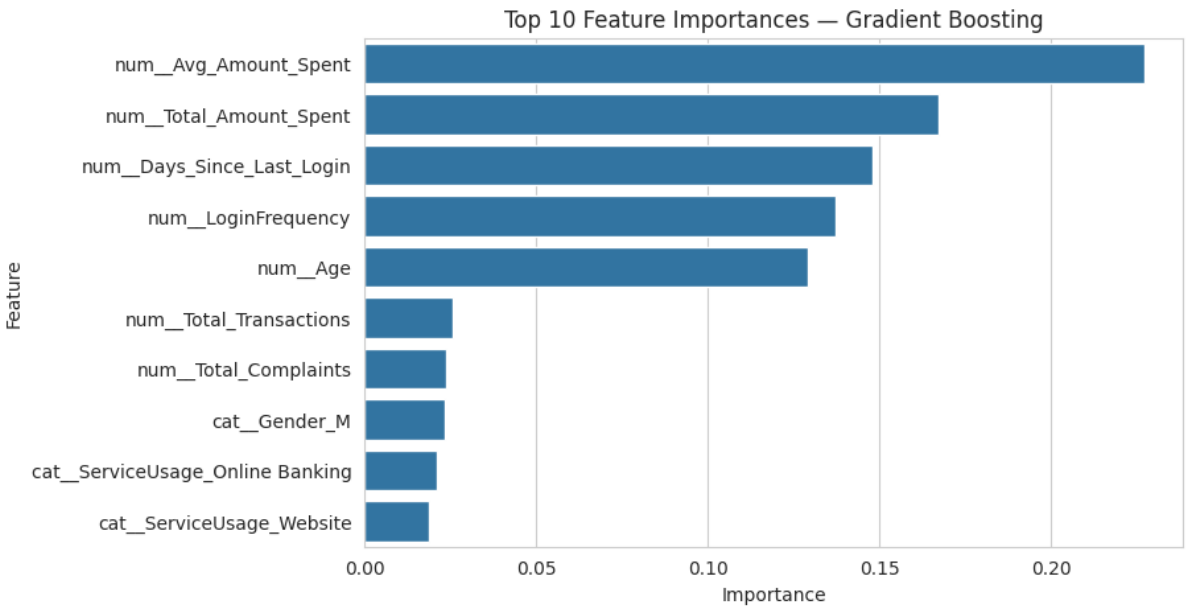
12. Feature Importance Analysis

Feature importance from the tuned Gradient Boosting model showed that customer engagement and spending behavior were the strongest predictors of churn.

Top influential features included:

- Average Amount Spent
- Total Amount Spent
- Days Since Last Login
- Login Frequency
- Age

Behavioural engagement metrics were more predictive than static demographic attributes.



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13. Key Findings

The analysis produced several important insights:

- Predictive performance was modest, with best ROC-AUC ≈ 0.57
- Non-linear ensemble models outperformed linear models
- Class imbalance handling alone did not significantly improve performance
- Customer engagement behavior was the strongest churn indicator
- Limited dataset size and feature richness constrained model performance

14. Business Implications

Even with moderate predictive power, churn models can provide business value when used to prioritize high-risk customers for retention campaigns.

Organizations should focus on:

- Monitoring declines in engagement and transaction behavior
- Identifying customers with reduced activity patterns
- Combining model predictions with domain knowledge and business rules

Improved data quality and richer behavioral features would likely yield substantial performance gains.

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15. Conclusion

This project demonstrated a complete machine learning workflow for customer churn prediction, including:

- Data preprocessing using pipelines
- Handling class imbalance
- Model comparison across multiple algorithms
- Hyperparameter optimization
- Feature importance interpretation

While predictive performance was constrained by limited feature signal, the modeling framework provides a strong foundation for real-world deployment and future improvement.